



The assessment of automated rating of L2 Mandarin prosody in lexical tone recognition and pauses

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Abstract

Providing feedback on spoken language tasks usually requires lots of efforts from language instructors. The use of automated engines can be helpful in rating pronunciations, especially with tonal languages like Mandarin. This study focused on evaluating prominent APIs for their abilities to assess oral readings by L2 Mandarin learners, with particular emphasis on detecting lexical tones and pauses.

In evaluating the recognition of lexical tones, the automated engines demonstrated the capacity to detect tones and provide a corresponding tonal pronunciation score. The study found that the overall accuracy of tonal diagnosis reached 80%, as determined through the calculation of false rejection and false acceptance rates. Furthermore, the rating distribution of the four lexical tones closely aligned with human ratings, thereby offering valuable insights for instructional strategies aimed at learners.

Regarding the assessment of intonation and other prosodic features, the automated engines were able to generate scores for pauses and overall fluency. It was observed that these scores exhibited an average correlation of 0.6 with human raters' detection of unnatural prosodic boundaries. As a result, the study recommends the inclusion of more detailed descriptions of prosodic features in current Mandarin automated rating models to enrich feedback and learning opportunities for learners.

Index Terms: L2 Mandarin, automated rating, lexical tones

1. Introduction

When assessing speaking skills in L2 learners, various task types such as reading aloud, presentations, dialogues, etc., are employed [1]. However, pronunciation undeniably remains a fundamental skill and a crucial descriptor in speaking rating criteria within second language learning [2]. Established marking descriptors typically categorize pronunciation into accuracy and fluency [3]. Accuracy includes the pronunciation of words, syllables, or phonemes, while fluency typically addresses aspects such as intonation, pace, rhythm, etc. [2]. This categorization can yield three scores: one for accuracy, another for fluency, and a third for an overall mark. Additionally, feedback comments provided to L2 learners hold significant value as they can pinpoint specific areas for improvement, such as repeated pronunciation errors or misplaced pauses [4]. Achieving accurate and fluent pronunciation can enhance learning confidence, as many L2 learners aspire to attain native-like proficiency [5].

However, a limited amount of classroom time can be allocated to pronunciation acquisition, as this time is often utilized for developing communicative tasks, grammar

exercises, and assessment practices, among other priorities [6]. Additionally, providing adequate feedback through human raters in classroom tests and standardized assessments has proven challenging for several reasons. Firstly, pronunciation rating can be time-consuming, requiring repeated playback of audio files to identify and highlight pronunciation issues. While this approach is beneficial for students, it is impractical to enumerate all errors [7]. Moreover, achieving inter-rater consistency among multiple markers for accuracy, fluency, and overall scores often proves difficult [8]. The advent of automated rating tools has emerged in language learning and testing since 2006, rapidly gaining popularity. For instance, organizations such as ETS, Duolingo, and Linguaskill have incorporated automated engines into their English language test products according to their published articles [7], [8], and [9].

Table 1: *Tones and tone sandhi in Mandarin*

Tone	Description
Tone 1	High level
Tone 2	High rising
Tone 3	Low falling-rising
Tone 4	High falling
Tone 0	Reduced
Tone 3 Sandhi	When Two Tone 3 in a row, the first changes to Tone 2
不 (meaning: No) Sandhi	不 pronounced as Tone 2 before Tone 4, as Tone 4 before Tone 1, 2, 3.
一 (meaning: One) Sandhi	一 pronounced as Tone 2 before Tone 4, as Tone 4 before Tone 1, 2, and 3.

In Mandarin learning and testing, pronunciation presents distinct challenges for L2 learners. Notably, among the top 10 languages by population, Mandarin stands alone as a tonal language, where semantic nuances are conveyed through lexical tones in addition to consonants and vowels [10]. Research indicates that the majority of pronunciation errors made by L2 Mandarin learners pertain to tones rather than vowels or consonants [11]. Each Mandarin character is associated with a specific lexical tone, as outlined in Table 2 [12]. Moreover, tone sandhi rules apply when pronouncing consecutive third tones, and specific tone pairs, such as those for the words “no” and “one” [13]. Linguists frequently explore lexical tones and sentence intonation concurrently, looking into the relationship of the two [14]. Regarding the role of automated ratings in Mandarin language learning, China has widely embraced this technological advancement. A recent report indicates that 19% of AI-related research were generated

from China [15]. Beyond the English language applications developed domestically, automated engines are used in China's national Mandarin speaking test. K-12 students also utilize automated engines for reading aloud exercises. Recently, mobile applications targeting L2 Mandarin learners have emerged [16]. However, unlike the automated rating systems available for English oral assessments, most Mandarin automated engines can only rate reading aloud tasks instead of the free constructed responses [17]. Given the limited research comparing automated engines with human raters for L2 Mandarin assessments, this article evaluates two prominent automated engines, assessing audio recordings from L2 Mandarin learners and comparing the results with human rater evaluations in answering the research question: what are the gaps of the two automated rating systems compare to human raters in terms of accuracy of specific tones and overall pronunciation scores among non-native Mandarin speakers?

2. Method

2.1. Participants and prompts

The recordings were collected from an online elementary L2 Mandarin course comprising 12 students, evenly divided with six male and six female students. Among them, eight students speak Indonesian as their first language, two speak English, one speaks Japanese, and one speaks Nepali.

All ten students whose first language is not English demonstrate fluency in English. Additionally, the eight Indonesian students exhibit a more proficient command of Cantonese or Hakka dialects compared to their limited proficiency in Mandarin.

The reading-aloud prompts were designed based on class materials encompassing four topics: weather, transportation, health, and travel, resulting in a total of 15 short sentences containing 145 characters. Among these sentences, two are interrogative, while the remainder are declarative. Students recorded their readings directly through a website equipped with the automated engine. Subsequently, these recordings were downloaded and assessed using a second automated model and human raters. The analysis includes a total of 146 recording entries and 1,388 syllables. The study covered all tone categories and the detailed breakdown numbers are available in Table 2. In addition, the pronunciation transcription, also called Pinyin is provided during the recording process.

Table 2: *Number of syllables with different tones and tone sandhi*

Tone 1	Tone 2	Tone 3	Tone 4	Tone 0	Tone Sandhi
20	22	33	49	20	7

2.2. Statistical analysis

In the framework proposed for evaluating and utilizing automated scoring, agreement with human scores and standardized mean score differences are defined as corresponding inferences within an argument-based validity alignment [18]. Several well-established English and Mandarin standardized tests have published their evaluation results based on this principle. The statistical methods utilized for comparing descriptors of accuracy, fluency, and total scores in this study are consistent with common approaches from prior research,

including means, standard deviation, and Pearson correlation. Box plots are also used for visual presentations illustrating score distributions [8], [9], [19].

The correlations between SpeechRater, the automated engine developed by the Educational Testing Service (ETS), and human raters for ten sets of speaking items range from 0.69 to 0.81, averaging 0.76. Two human raters exhibited correlations ranging from 0.69 to 0.76, given that the second human rater evaluated only 10% of the data assessed by the first human rater [8]. Meanwhile, correlations among five human raters in the Mandarin Proficiency Test for National Minorities ranged from 0.84 to 0.90, with the automated rater achieving a correlation as high as 0.93 when compared to the average human rater score [19]. Another study focusing on the custom automated speech engine (CASE) for the Cambridge Assessment English Linguaskill General Speaking test also recommended using the average human rater scores as the gold standard criterion for evaluating automated ratings [9]. It is noteworthy that in the three aforementioned studies, the automated engines employed were customized for specific test purposes. However, given this study's focus on assessing the reliability of existing automated services to be applied in regular classroom settings, the selected automated engines can not be custom-made. Furthermore, the scales used in the three studies differ from the 0 to 100 scale for classroom settings, instead utilizing a condensed scale ranging from 0 to 6 or less for larger standardized tests. Consequently, the mean differences observed in prior studies are not listed for comparison.

For detailed tone recognition and error diagnosis in Mandarin, three metrics were proposed before: False Rejection Rate (FRR), representing the percentage of correctly pronounced units falsely rejected; False Acceptance Rate (FAR), denoting the percentage of mispronounced units falsely accepted; and Diagnostic Accuracy (DA), indicating the percentage of phones correctly recognized relative to human raters. The average DA across the four tones (Tones 1 to 4) has reached 82.45% for the development of automated engines [20]. This study will also calculations for FRR, FAR, and DA across the four tones, as well as the Tone 0, tonal sandhi, and pauses.

2.3. Rating process

2.3.1 Human raters

Two human raters with over ten years of teaching experience participated in the rating process. Prior to the rating, two raters discussed standards for assessing accuracy and fluency. Notes were included in the marking sheets to capture descriptions related to fluency, such as pronounced hesitations, pauses, chopiness, and incoherence. Following the initial round of assessments, inter-rater consistency was evaluated, resulting in an initial correlation of 0.65 between the human raters. For audio samples with discrepancies exceeding 10% in human raters' evaluations, followed by analysis, a comprehensive remarking was carried out with the 146 entries of audios. The discrepancies mainly focus on accents from some Indonesian learners similar to authentic Mandarin accents from dialects speakers. After the remarking, the human raters achieved a second correlation of 0.75. The average scores assigned by these two human raters were then established as the gold standard, listed as H in the following section of results. This is the value which the performance of automated engines would be benchmarked with.

For error diagnosis, one of the human raters meticulously assessed the initial, final, and tonal accuracy of all 1,388 syllables. The second human rater selected 15% of all syllables found no discrepancies between their evaluations.

2.3.2 Automated raters

Two automated engines were selected for this project: one developed by a Chinese company, referred to as A1, and another developed by an international company, referred as A2. These automated raters equipped different rating models for characters, words, sentences, and passages. This article specifically focuses on the model of sentences. During the rating process, the text of the speech must be uploaded concurrently with the speech recording. Engine A1 offers a broader spectrum of returned results compared to A2. Specifically, A1 provides metrics such as overall score; completeness; accuracy score with sub categories of pronunciation without tone, tone score, tone projection, and syllable score; fluency score with sub categories in number of pauses and speech rate. Pauses is the only sub category in fluency selected for this phase of the study. In contrast, A2 currently only offers an overall pronunciation score, accuracy score, fluency score, and completeness for Mandarin assessments. Therefore, the comparative analysis of scores is conducted among A1, A2, and the mean of the human raters as H. Meanwhile, error diagnosis pertaining to the five tones, tone sandhi, and pauses is specifically examined between A1 and H for further evaluation.

3. Results

3.1. Comparison of scores in lexical accuracies and fluencies

As the pronunciation transcriptions are provided during the read-aloud tasks, minimal pronunciation errors with vowels and consonants were observed across all 1,388 syllables, indicated by the assessments from both human raters. As a result, accuracy scores can be interpreted as measures of lexical tone accuracy. As outlined in Table 3, the mean scores for both overall and accuracy ratings between human raters and the two automated systems are relatively consistent, with the most significant discrepancies being 4 points (between H and A1) and 5 points (between H and A2). However, notable variations emerged in fluency scores, with discrepancies reaching 30.4 points (between H and A1) and 14.1 points (between H and A2). A closer examination of the standard deviations (SD) presented in Table 3 reveals that human raters tend to assign consistent scores for both accuracy and fluency, as evidenced by minimal fluctuations in their scores. In contrast, A1 exhibits stringent criteria for fluency ratings, registering a mean score of 53.1, while A2 demonstrates leniency with a mean fluency score of 97.9. This discrepancy may be attributed to A1's incorporation of multiple sub-categories within fluency assessment, such as pauses and speech rate, whereas A2 offers a singular fluency score.

Table 3: Means and standard deviations for the human rater score and the two automated engines

	H	A1	A2
Mean - Accuracy	85.1	89	80.1
Mean - Fluency	83.8	53.1	97.9
Mean - Overall	84.7	80.7	82.9

SD - Accuracy	9.4	5.4	10.7
SD - Fluency	9.9	18.8	4.5
SD - Overall	9.6	6.9	11.9

Table 4 presents the correlation scores among the three raters. The highest correlation, at 0.60, is observed between the scores of the human raters and A1, which is slightly below the lowest correlation of 0.69 reported in prior studies. While the H-A2 correlation in accuracy surpasses that of H-A1 by a marginal extent, the H-A1 correlations in fluency and overall scores are notably higher. This suggests that A1 aligns more closely with the established human standard. Furthermore, given that A1 offers more granular feedback for error diagnosis, it is conceivable that incorporating sub-metrics within accuracy and fluency assessments for automated raters could enhance alignment with human raters and provide more constructive feedback for language learners. Additionally, the correlation scores between the two automated raters consistently fall below those between human and automated raters, underscoring distinct developmental approaches across different automated models.

Table 4: Correlations between the human rater and the two automated engines

	H-A1	H-A2	A1-A2
Correlation - Accuracy	0.31	0.39	0.34
Correlation - Fluency	0.60	0.20	0.17
Correlation - Overall	0.51	0.37	0.47

The three box plots presented below depict the distribution of scores across accuracy, fluency, and overall ratings. The box area represents the interquartile range, capturing 50% of the score distribution, while the whiskers above and below the box indicate potential skewness in the data.

In Figure 1, focusing on the overall scores, A2 notably exhibits a broader distribution of marks, including several exceptionally low results. Upon rechecks by human raters of the two lowest audio entries, no correlation was identified with the scores assigned by A2.

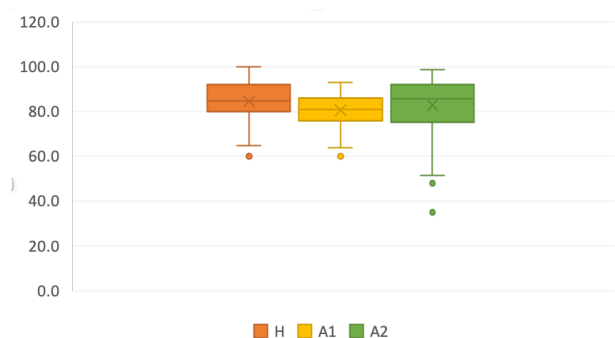


Figure 1: Box plots of overall scores of the human rater and the two automated engines

In Figure 2 for accuracies, A1 displayed a significantly narrower score distribution, with the lowest score notably higher than that of H. This suggests that A1 may not effectively identify low-performance readings, thereby potentially compromising its ability to provide reliable assessments for a sizable group of learners with varied proficiencies or individual learners transitioning from lower to higher proficiency levels.

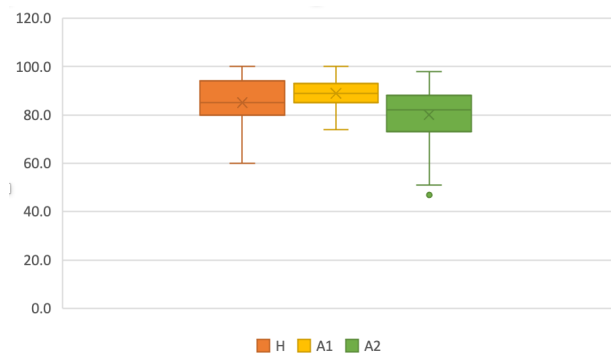


Figure 2: Box plots of scores of the human rater and the two automated engines in Accuracy

Similar to the trends observed in score comparisons, Figure 3, focusing on fluency, presents the most divergent outcomes, with A1 scoring much lower than H and A2 achieving nearly all top scores. This underscores the need for further research and development in automated scoring systems for fluency and related prosodic metrics.

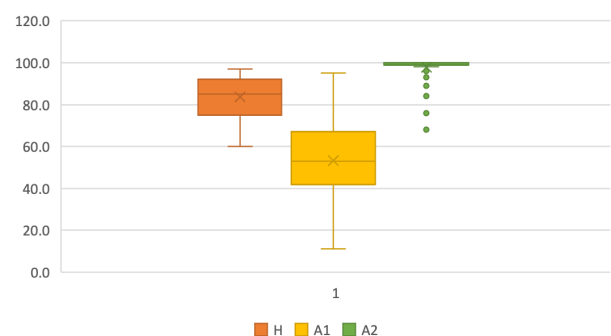


Figure 3: Box plots of scores of the human rater and the two automated engines in Fluency

3.2 Comparison of error detection and diagnosis

Table 5 summarizes the ratings for the five tones, tone sandhi, and the detection of pauses based on the definitions provided in Section 2.2. All Diagnostic Accuracy (DA) scores exceeded 70%, aligning more closely with findings from previous studies than the correlation results. Moreover, no significant differences were observed among the various tones, tone sandhi, and pauses. Error detection and diagnosis represent the most labor-intensive aspects of pronunciation rating, yet they are highly effective for automated raters. The positive results in the study indicating promising potential for deploying automated systems to offer specific feedback to learners.

Although the DA results from automated engines are more satisfactory than the marks they produced, the False Rejection

Rate (FRR) still exceeds the False Acceptance Rate (FAR) of a minimum 10%. This implies that a higher percentage of correctly pronounced items are falsely rejected compared to the mispronounced items being falsely accepted. Therefore, the automated model should be less stringent in identifying phoneme errors, thereby increasing tolerance for pronunciation variations among L2 learners. This approach will allow for a focused effort on addressing more severe errors.

Table 5: three metrics of different tones comparing A1 with H

	FRR	FAR	DA
Tone 1	25%	16%	75%
Tone 2	19%	8%	85%
Tone 3	30%	13%	84%
Tone 4	15%	7%	73%
Tone 0	22%	12%	78%
Tone Sandhi	13%	10%	81%
Pauses	22%	3%	76%

4. Discussion

Although automated raters have shown success in large-scale standardized language tests for English and Mandarin, this study did not find similar efficacy when utilizing these tools for L2 Mandarin learning within classroom settings. One possible reason for this discrepancy is that standardized tests employ customized automated models, ensuring greater reliability. Additionally, prior research on standardized tests often utilized a more limited scoring scale, typically six or below, whereas classroom assessments commonly use a 100-point scale. Thus, readily available automated tools may not offer significant advantages for language instructors focusing on accuracy and fluency assessment. This also demonstrates that while automated engines may effectively replace human raters in standardized tests, their applicability to L2 classrooms remains questionable.

However, if the selected automated engine can accurately detect specific tones and diagnose errors, they could serve as valuable tools for providing detailed feedback to L2 learners. Such feedback would allow students to systematically improve their pronunciation of individual tones, serving as a beneficial supplement to the more communicative driven classroom learning. Furthermore, automated engines offer the advantage of delivering feedback more quickly and in a personalized manner.

It is important to note that this study did not explore the relationship between speech rate in the fluency descriptor of automated models and the fluency scores assigned by human raters. To enhance alignment in fluency and prosody assessments, future research should develop and analyze more additional metrics in automated rating. Ideally, these metrics can also be sensible with both human raters and learners, incorporating elements such as visualization of pitch curves and intonation patterns. Furthermore, additional audio recordings with more evenly distributed L1 background will be collected to study whether this influences the accuracy of scoring. By doing so, automated technologies could become more effective in providing constructive feedback for improving L2 Mandarin prosody and pronunciation.

5. References

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